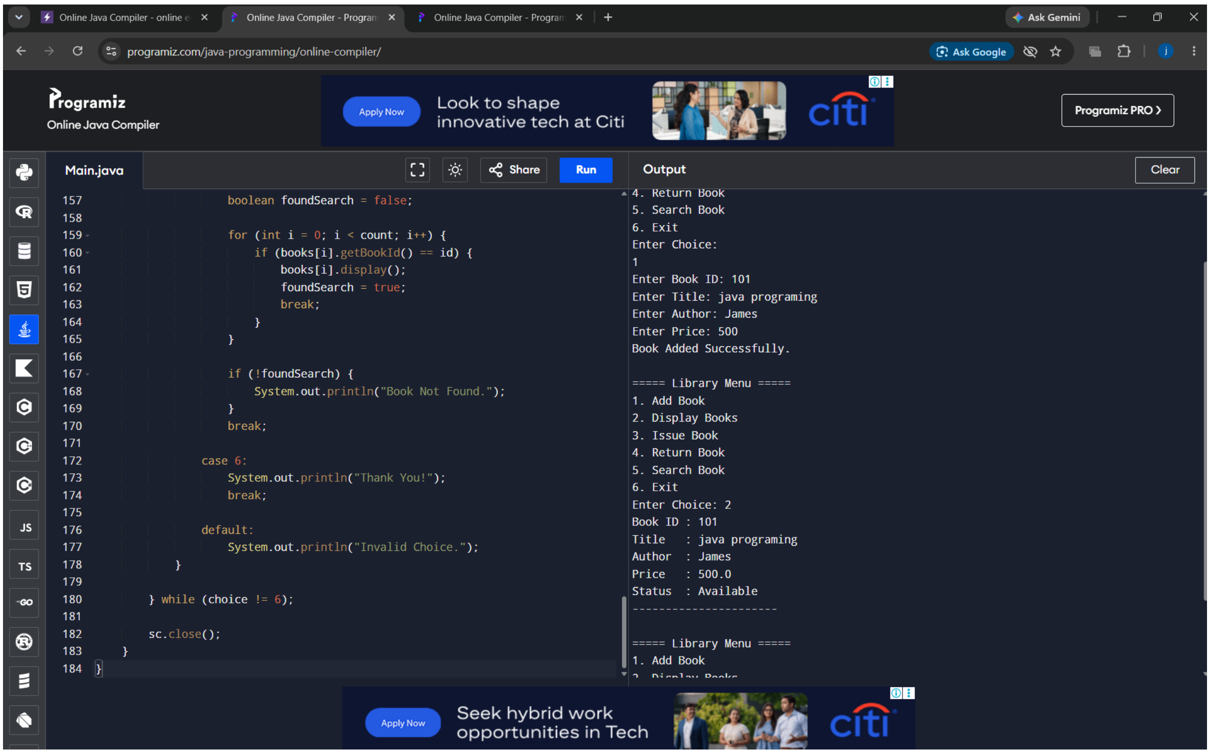
```java
import java.util.Scanner;

class Book {
    private int bookId;
    private String title;
    private String author;
    private double price;
    private boolean issued;
    Book(int id, String t, String a, double p) {
        bookId = id;
        title = t;
        author = a;
        price = p;
        issued = false;
    }
    void display() {
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Author  : " + author);
        System.out.println("Price   : " + price);
        System.out.println("Status  : " + (issued ? "Issued" : "Available"));
    }
    void issueBook() {
        if (!issued) {
            issued = true;
            System.out.println("Book Issued Successfully.");
        } else {
            System.out.println("Book Already Issued.");
```

```
        }
    }
    void returnBook() {
        if (issued) {
            issued = false;
            System.out.println("Book Returned Successfully.");
        } else {
            System.out.println("Book is Already Available.");
        }
    }
    int getBookId() {
        return bookId;
    }
}

public class LibraryManagement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Book[] books = new Book[5];
        int count = 0;
        int choice;
        do {
            System.out.println("\n----- Library Menu -----");
            System.out.println("1. Add Book");
            System.out.println("2. Display Books");
            System.out.println("3. Issue Book");
            System.out.println("4. Return Book");
            System.out.println("5. Search Book");
```

```java
System.out.println("6. Exit");
System.out.print("Enter Choice: ");
choice = sc.nextInt();
switch (choice) {
  case 1:
    System.out.print("Enter Book ID: ");
    int id = sc.nextInt();
    sc.nextLine();
    System.out.print("Enter Title: ");
    String title = sc.nextLine();
    System.out.print("Enter Author: ");
    String author = sc.nextLine();
    System.out.print("Enter Price: ");
    double price = sc.nextDouble();
    books[count] = new Book(id, title, author, price);
    count++;
    System.out.println("Book Added Successfully.");
    break;
  case 2:
    for (int i = 0; i < count; i++) {
      books[i].display();
      System.out.println();
    }
    break;
  case 3:
    System.out.print("Enter Book ID: ");
    id = sc.nextInt();
    for (int i = 0; i < count; i++) {
```

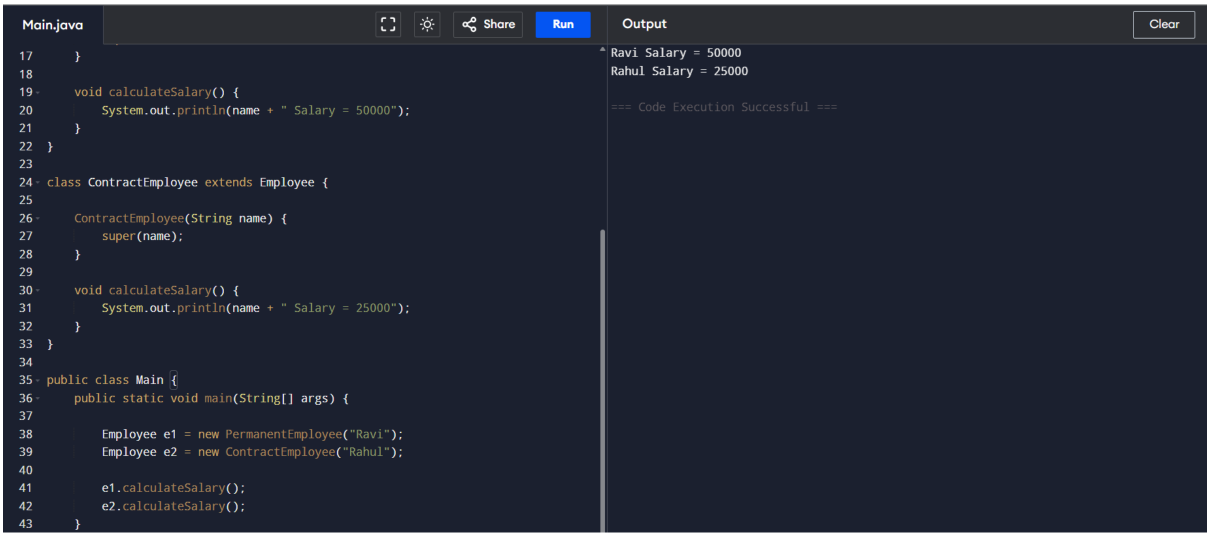
```
            if (books[i].getBookId() == id) {
                books[i].issueBook();
            }
        }
        break;
    case 4:
        System.out.print("Enter Book ID: ");
        id = sc.nextInt();
        for (int i = 0; i < count; i++) {
            if (books[i].getBookId() == id) {
                books[i].returnBook();
            }
        }
        break;
    case 5:
        System.out.print("Enter Book ID: ");
        id = sc.nextInt();
        boolean found = false;
        for (int i = 0; i < count; i++) {
            if (books[i].getBookId() == id) {
                books[i].display();
                found = true;
            }
        }
        if (!found) {
            System.out.println("Book Not Found.");
        }
        break;
```

```
            case 6:
                System.out.println("Thank You!");
                break;
            default:
                System.out.println("Invalid Choice.");
        }
    } while (choice != 6);
    sc.close();
  }
}
```



```
2  class Employee {
    String name;
    Employee(String name) {
        this.name = name;
    }
    void calculateSalary() {
        System.out.println("Salary");
    }
}
class PermanentEmployee extends Employee {
    PermanentEmployee(String name) {
        super(name);
    }
    void calculateSalary() {
        System.out.println(name + " Salary = 50000");
    }
}
class ContractEmployee extends Employee {
    ContractEmployee(String name) {
        super(name);
    }
    void calculateSalary() {
        System.out.println(name + " Salary = 25000");
    }
}
public class Main {
    public static void main(String[] args) {
        Employee e1 = new PermanentEmployee("Ravi");
```

```
        Employee e2 = new ContractEmployee("Rahul");

        e1.calculateSalary();

        e2.calculateSalary();

    }

}
```

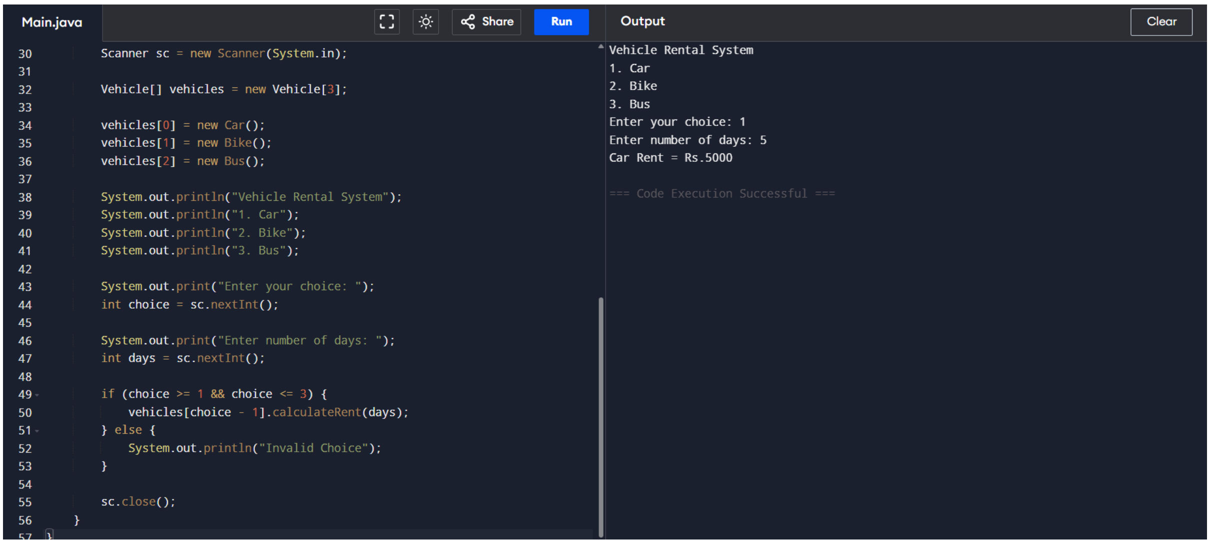


3.

```
import java.util.Scanner;


class Vehicle {

    void calculateRent(int days) {

        System.out.println("Vehicle Rent");

    }

}


class Car extends Vehicle {

    void calculateRent(int days) {

        System.out.println("Car Rent = Rs." + (days * 1000));

    }

}
```

```
class Bike extends Vehicle {
    void calculateRent(int days) {
        System.out.println("Bike Rent = Rs." + (days * 300));
    }
}

class Bus extends Vehicle {
    void calculateRent(int days) {
        System.out.println("Bus Rent = Rs." + (days * 2000));
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Vehicle[] vehicles = new Vehicle[3];

        vehicles[0] = new Car();
        vehicles[1] = new Bike();
        vehicles[2] = new Bus();

        System.out.println("Vehicle Rental System");
        System.out.println("1. Car");
        System.out.println("2. Bike");
        System.out.println("3. Bus");
```

```java
        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();

        System.out.print("Enter number of days: ");
        int days = sc.nextInt();

        if (choice >= 1 && choice <= 3) {
            vehicles[choice - 1].calculateRent(days);
        } else {
            System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```



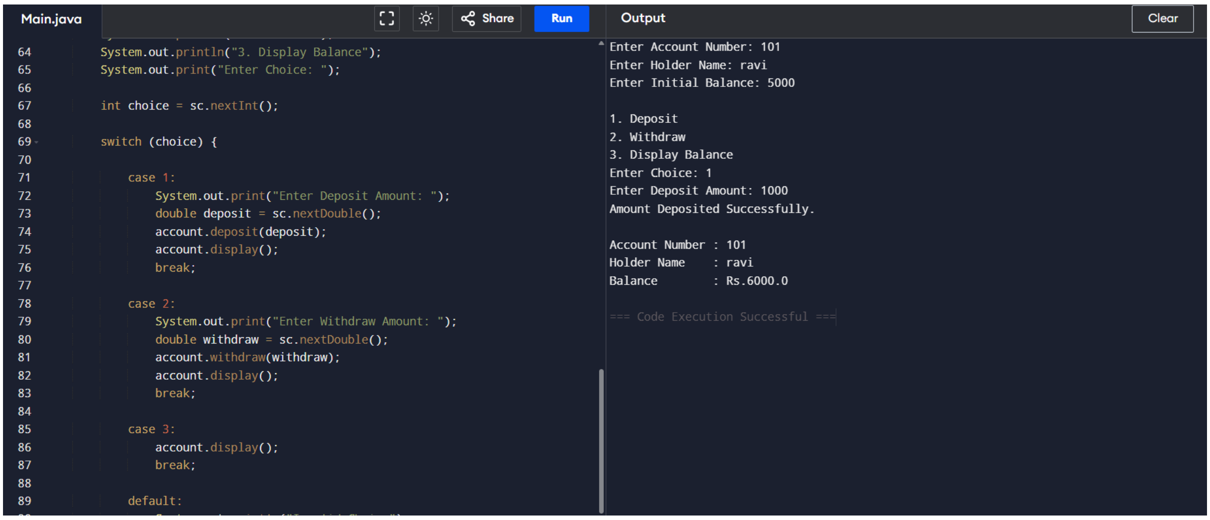
4. import java.util.Scanner;

```
class BankAccount {

   // Private data members (Data Hiding)
   private int accountNumber;
   private String holderName;
   private double balance;

   // Constructor
   BankAccount(int accNo, String name, double bal) {
      accountNumber = accNo;
      holderName = name;
      balance = bal;
   }

   // Deposit method
   void deposit(double amount) {
      if (amount > 0) {
         balance = balance + amount;
         System.out.println("Amount Deposited Successfully.");
      } else {
         System.out.println("Invalid Deposit Amount.");
      }
   }

   // Withdraw method
   void withdraw(double amount) {
```

```
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Amount Withdrawn Successfully.");
        } else {
            System.out.println("Insufficient Balance.");
        }
    }

    // Display Account Details
    void display() {
        System.out.println("\nAccount Number : " + accountNumber);
        System.out.println("Holder Name    : " + holderName);
        System.out.println("Balance        : Rs." + balance);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Account Number: ");
        int accNo = sc.nextInt();
        sc.nextLine();

        System.out.print("Enter Holder Name: ");
        String name = sc.nextLine();
```

```java
System.out.print("Enter Initial Balance: ");
double balance = sc.nextDouble();

BankAccount account = new BankAccount(accNo, name, balance);

System.out.println("\n1. Deposit");
System.out.println("2. Withdraw");
System.out.println("3. Display Balance");
System.out.print("Enter Choice: ");

int choice = sc.nextInt();

switch (choice) {

  case 1:
    System.out.print("Enter Deposit Amount: ");
    double deposit = sc.nextDouble();
    account.deposit(deposit);
    account.display();
    break;

  case 2:
    System.out.print("Enter Withdraw Amount: ");
    double withdraw = sc.nextDouble();
    account.withdraw(withdraw);
    account.display();
    break;
```

```
            case 3:
                account.display();
                break;

            default:
                System.out.println("Invalid Choice");
        }

        sc.close();
    }
}
```

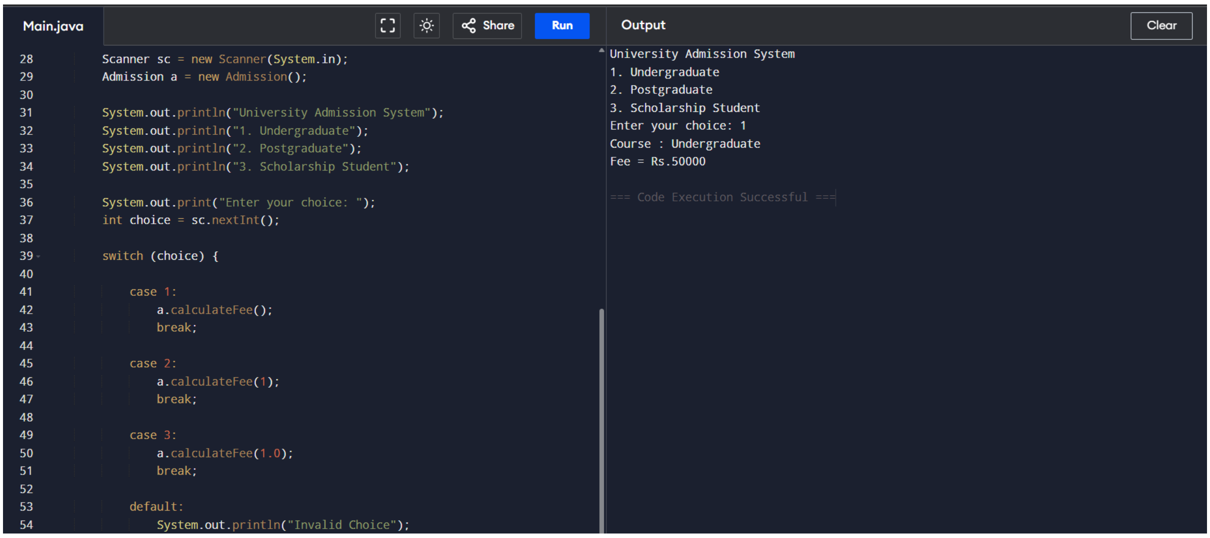


5. import java.util.Scanner;

```
class Admission {

    // Undergraduate Fee
    void calculateFee() {
        System.out.println("Course : Undergraduate");
        System.out.println("Fee = Rs.50000");
    }

    // Postgraduate Fee
    void calculateFee(int pg) {
        System.out.println("Course : Postgraduate");
        System.out.println("Fee = Rs.70000");
    }

    // Scholarship Student Fee
    void calculateFee(double scholarship) {
        System.out.println("Course : Scholarship");
        System.out.println("Fee = Rs.20000");
    }
}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

```
    Admission a = new Admission();

    System.out.println("University Admission System");
    System.out.println("1. Undergraduate");
    System.out.println("2. Postgraduate");
    System.out.println("3. Scholarship Student");

    System.out.print("Enter your choice: ");
    int choice = sc.nextInt();

    switch (choice) {

        case 1:
            a.calculateFee();
            break;

        case 2:
            a.calculateFee(1);
            break;
        case 3:
            a.calculateFee(1.0);
            break;
        default:
            System.out.println("Invalid Choice");
    }

    sc.close();
}
```

}

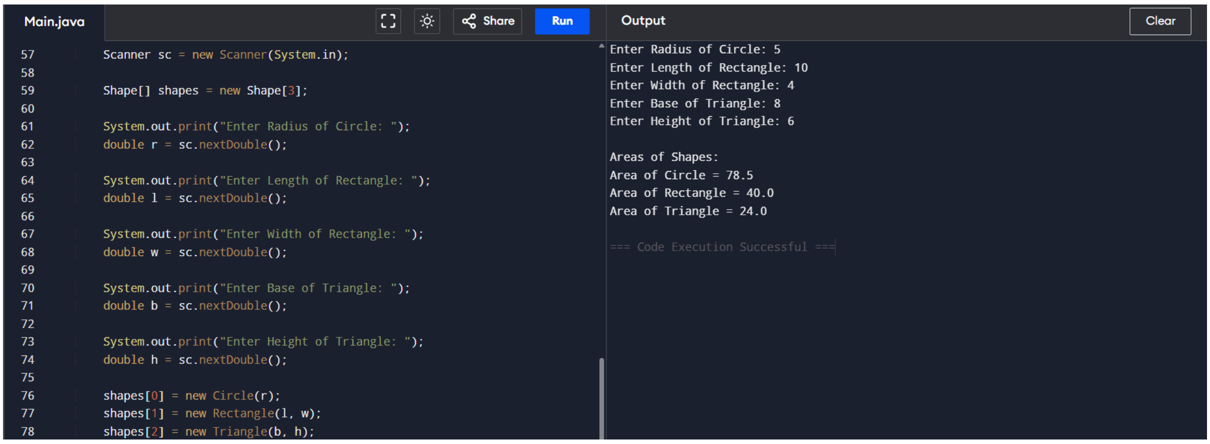


6. import java.util.Scanner;

```
abstract class Shape {
   abstract void area();
}
class Circle extends Shape {
   double radius;
   Circle(double r) {
      radius = r;
   }
   void area() {
      double a = 3.14 * radius * radius;
      System.out.println("Area of Circle = " + a);
   }
}
class Rectangle extends Shape {
   double length, width;
   Rectangle(double l, double w) {
      length = l;
```

```
        width = w;
    }

    void area() {
        double a = length * width;
        System.out.println("Area of Rectangle = " + a);
    }
}
class Triangle extends Shape {
    double base, height;
    Triangle(double b, double h) {
        base = b;
        height = h;
    }
    void area() {
        double a = 0.5 * base * height;
        System.out.println("Area of Triangle = " + a);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Shape[] shapes = new Shape[3];
        System.out.print("Enter Radius of Circle: ");
        double r = sc.nextDouble();
        System.out.print("Enter Length of Rectangle: ");
        double l = sc.nextDouble();
```

```java
        System.out.print("Enter Width of Rectangle: ");
        double w = sc.nextDouble();
        System.out.print("Enter Base of Triangle: ");
        double b = sc.nextDouble();
        System.out.print("Enter Height of Triangle: ");
        double h = sc.nextDouble();
        shapes[0] = new Circle(r);
        shapes[1] = new Rectangle(l, w);
        shapes[2] = new Triangle(b, h);
        System.out.println("\nAreas of Shapes:");
        for (Shape s : shapes) {
            s.area();
        }
        sc.close();
    }
}
```

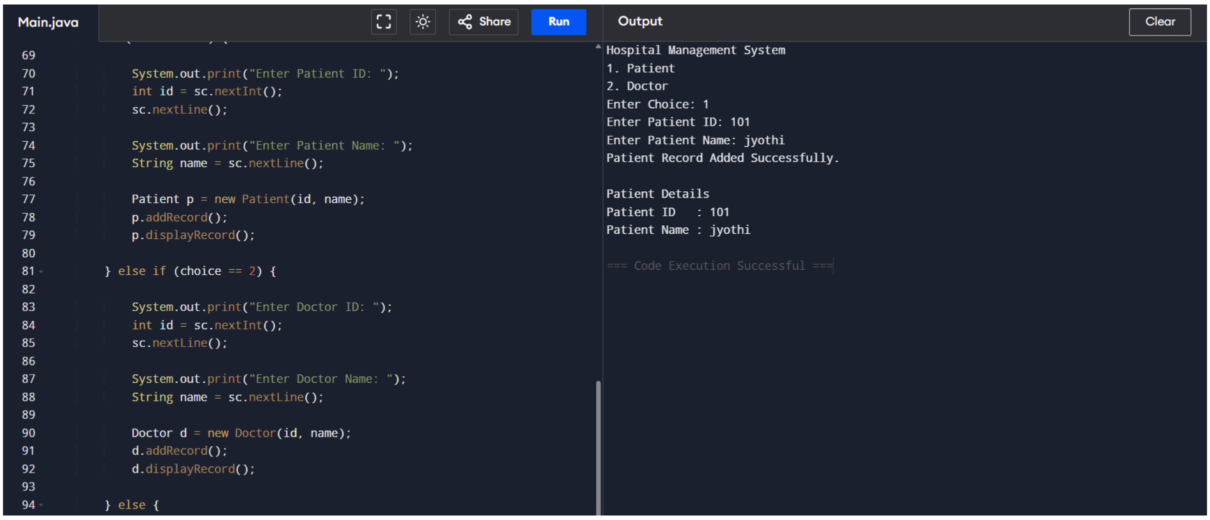


7. import java.util.Scanner;

```
interface MedicalRecord {
    void addRecord();
    void displayRecord();
}
class Patient implements MedicalRecord {
    int patientId;
    String patientName;
    Patient(int id, String name) {
        patientId = id;
        patientName = name;
    }
    public void addRecord() {
        System.out.println("Patient Record Added Successfully.");
    }

    public void displayRecord() {
        System.out.println("\nPatient Details");
        System.out.println("Patient ID   : " + patientId);
        System.out.println("Patient Name : " + patientName);
    }
}
class Doctor implements MedicalRecord {
    int doctorId;
    String doctorName;
    Doctor(int id, String name) {
        doctorId = id;
        doctorName = name;
```

```
    }


    public void addRecord() {
        System.out.println("Doctor Record Added Successfully.");
    }
    public void displayRecord() {
        System.out.println("\nDoctor Details");
        System.out.println("Doctor ID   : " + doctorId);
        System.out.println("Doctor Name : " + doctorName);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Hospital Management System");
        System.out.println("1. Patient");
        System.out.println("2. Doctor");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();
        sc.nextLine();
        if (choice == 1) {
            System.out.print("Enter Patient ID: ");
            int id = sc.nextInt();
            sc.nextLine();
            System.out.print("Enter Patient Name: ");
            String name = sc.nextLine();
            Patient p = new Patient(id, name);
            p.addRecord();
```

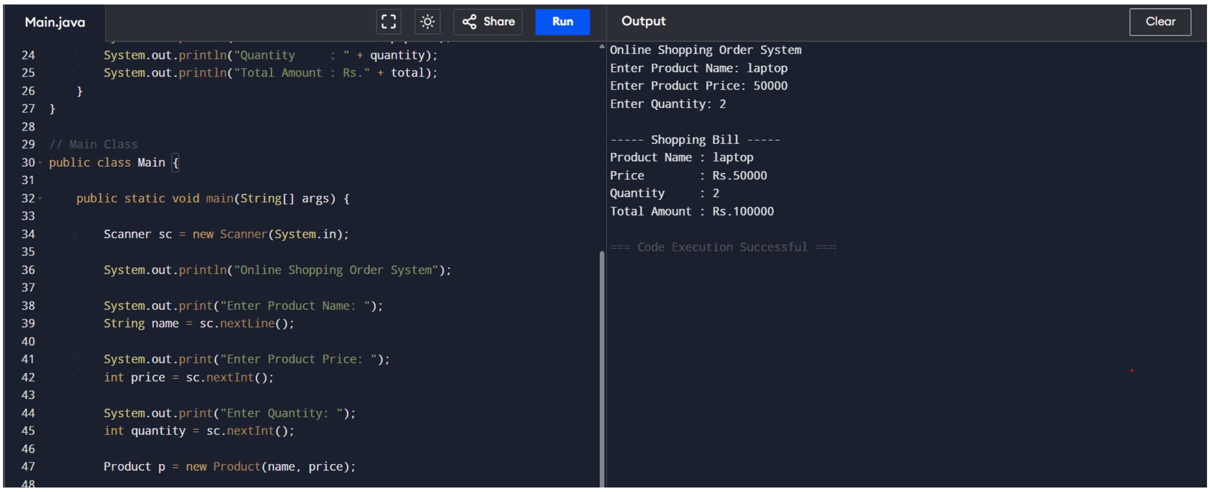
```
            p.displayRecord();

        } else if (choice == 2) {

            System.out.print("Enter Doctor ID: ");

            int id = sc.nextInt();

            sc.nextLine();

            System.out.print("Enter Doctor Name: ");

            String name = sc.nextLine();

            Doctor d = new Doctor(id, name);

            d.addRecord();

            d.displayRecord();

        } else {

            System.out.println("Invalid Choice");

        }

        sc.close();

    }

}
```



8. import java.util.Scanner;

```
class Product {
    String productName;
    int price;

    Product(String name, int price) {
        this.productName = name;
        this.price = price;
    }
}
class Order {
    void placeOrder(Product p, int quantity) {
        int total = p.price * quantity;
        System.out.println("\n----- Shopping Bill -----");
        System.out.println("Product Name : " + p.productName);
        System.out.println("Price        : Rs." + p.price);
        System.out.println("Quantity     : " + quantity);
        System.out.println("Total Amount : Rs." + total);
    }
}
public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("Online Shopping Order System");
```

```java
        System.out.print("Enter Product Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Product Price: ");
        int price = sc.nextInt();

        System.out.print("Enter Quantity: ");
        int quantity = sc.nextInt();

        Product p = new Product(name, price);

        Order o = new Order();

        o.placeOrder(p, quantity);

        sc.close();
    }
}
```

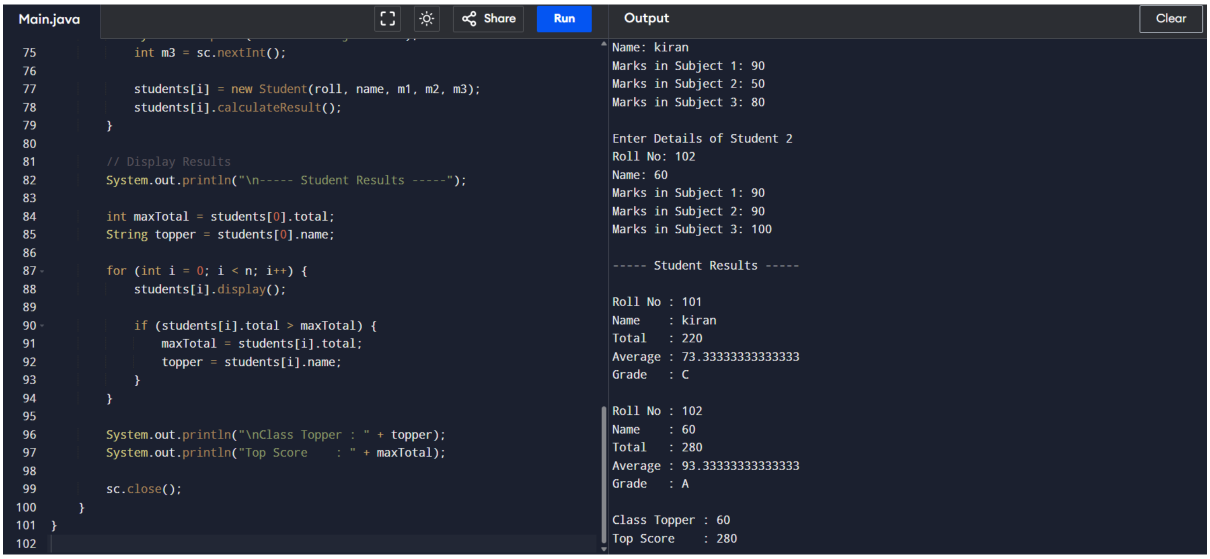


9. import java.util.Scanner;

```
class Student {
    int rollNo;
    String name;
    int m1, m2, m3;
    int total;
    double average;
    String grade;
    Student(int rollNo, String name, int m1, int m2, int m3) {
        this.rollNo = rollNo;
        this.name = name;
        this.m1 = m1;
        this.m2 = m2;
        this.m3 = m3;
    }
    void calculateResult() {
        total = m1 + m2 + m3;
        average = total / 3.0;

        if (average >= 90)
            grade = "A";
        else if (average >= 75)
            grade = "B";
        else if (average >= 60)
            grade = "C";
        else
            grade = "Fail";
```

```java
    }
    void display() {
        System.out.println("\nRoll No : " + rollNo);
        System.out.println("Name    : " + name);
        System.out.println("Total   : " + total);
        System.out.println("Average : " + average);
        System.out.println("Grade   : " + grade);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Number of Students: ");
        int n = sc.nextInt();
        Student[] students = new Student[n];
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter Details of Student " + (i + 1));
            System.out.print("Roll No: ");
            int roll = sc.nextInt();
            sc.nextLine();
            System.out.print("Name: ");
            String name = sc.nextLine();
            System.out.print("Marks in Subject 1: ");
            int m1 = sc.nextInt();
            System.out.print("Marks in Subject 2: ");
            int m2 = sc.nextInt();
            System.out.print("Marks in Subject 3: ");
```

```
            int m3 = sc.nextInt();


            students[i] = new Student(roll, name, m1, m2, m3);
            students[i].calculateResult();
        }
        System.out.println("\n----- Student Results -----");
        int maxTotal = students[0].total;
        String topper = students[0].name;
        for (int i = 0; i < n; i++) {
            students[i].display();
            if (students[i].total > maxTotal) {
                maxTotal = students[i].total;
                topper = students[i].name;
            }
        }
        System.out.println("\nClass Topper : " + topper);
        System.out.println("Top Score    : " + maxTotal);
        sc.close();
    }
}
```

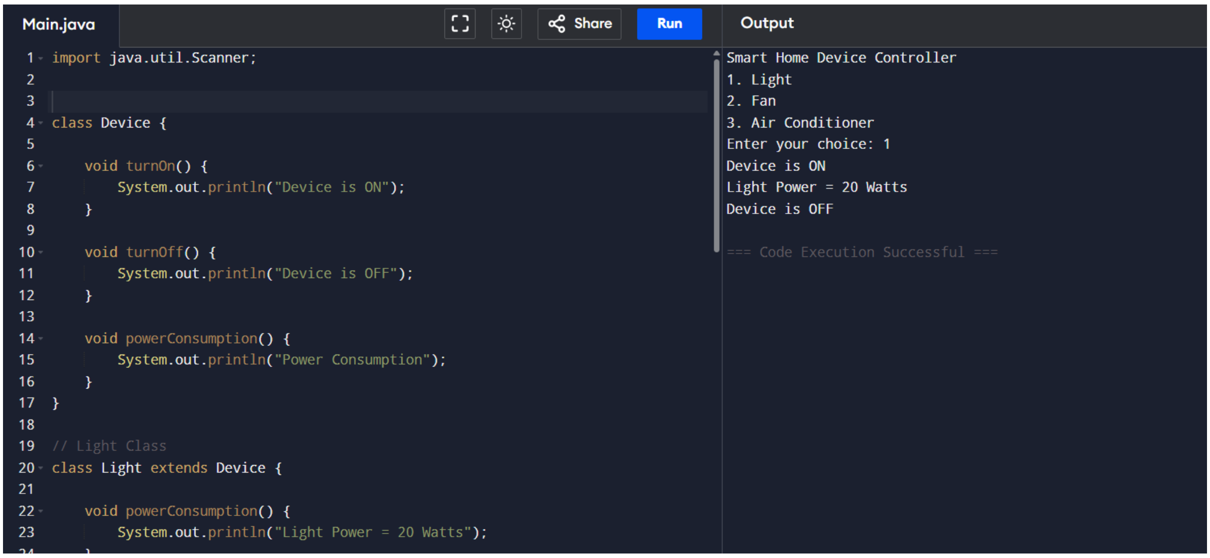


10. import java.util.Scanner;

```
class Device {
    void turnOn() {
        System.out.println("Device is ON");
    }
    void turnOff() {
        System.out.println("Device is OFF");
    }
    void displayPower() {
        System.out.println("Power Consumption");
    }
}
class Light extends Device {
    void displayPower() {
        System.out.println("Light Power Consumption = 20 Watts");
    }
}
class Fan extends Device {


    void displayPower() {
        System.out.println("Fan Power Consumption = 75 Watts");
    }
}
class AirConditioner extends Device {
    void displayPower() {
        System.out.println("Air Conditioner Power Consumption = 1500 Watts");
    }
}
```

```java
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Device device = null;
        System.out.println("===== Smart Home Device Controller =====");
        System.out.println("1. Light");
        System.out.println("2. Fan");
        System.out.println("3. Air Conditioner");
        System.out.print("Enter your choice: ");
        int choice = sc.nextInt();
        switch (choice) {
            case 1:
                device = new Light();
                break;
            case 2:
                device = new Fan();
                break;
            case 3:
                device = new AirConditioner();
                break;
            default:
                System.out.println("Invalid Choice");
                return;
        }
        device.turnOn();
        device.displayPower();
        device.turnOff();
```

```
        sc.close();
    }
}
```


Main.java
Share
Run
Output
import java.util.Scanner;
class Device {
void turnOn() {
System.out.println("Device is ON");
}
void turnOff() {
System.out.println("Device is OFF");
}
void powerConsumption() {
System.out.println("Power Consumption");
}
}
// Light Class
class Light extends Device {
void powerConsumption() {
System.out.println("Light Power = 20 Watts");
Smart Home Device Controller
1. Light
2. Fan
3. Air Conditioner
Enter your choice: 1
Device is ON
Light Power = 20 Watts
Device is OFF
=== Code Execution Successful ===